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Section-B

Course:B.tech Cse Core

Assignment 4 :

❖ Aim

To design and develop a full-stack To-Do List web application using HTML, CSS, JavaScript, Python, and Flask, enabling users to add, edit, delete, mark complete, and manage tasks efficiently.

❖ Tasks (Flask Version)

• Set up a Flask project structure
• Create a Task model using a database (SQLite with SQLAlchemy)
• Design HTML templates using Jinja2
• Implement backend logic using Flask routes
• Create REST API endpoints (GET, POST, PUT, DELETE)
• Use JavaScript Fetch API for dynamic updates (no reload)
• Add task operations (create, update, delete, toggle)
• Implement filtering (All, Active, Completed)

❖ Technologies Used

• Frontend: HTML5, CSS3, JavaScript (ES6)
• Backend: Python 3
• Framework: Flask
• Database: SQLite (via SQLAlchemy)
• API Handling: Fetch API (AJAX)

❖ Code :

❯ HTML CODE :

❖ base.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>{% block title %}Flask ToDo App{% endblock %}</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link
    href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&fa
    mily=DM+Sans:opsz,wght@9..40,400;9..40,500;9..40,600;9..40,700&display=swap"
    rel="stylesheet">
<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
    awesome/6.4.0/css/all.min.css">
</head>
<body>
<div class="container">
<header>
<h1><i class="fas fa-check-circle"></i> My ToDo List</h1>
<p class="subtitle">Stay organized, stay productive</p>
</header>

{% with messages = get_flashed_messages(with_categories=true) %}
{% if messages %}
{% for category, message in messages %}
<div class="flash-message flash-{{ category }}">{{ message }}</div>
{% endfor %}
{% endif %}
{% endwith %}

{% block content %}{% endblock %}
</div>

<script src="{{ url_for('static', filename='js/main.js') }}"></script>
</body>
</html>
```

❖ **index.html:**

```
{% extends "base.html" %}
{% block title %}My ToDo List — Flask App{% endblock %}

{% block content %}

<!-- ═══════════════════════════════════════
ADD TASK SECTION
═══════════════════════════════════════ -->
<section class="add-todo-section">
<form action="{{ url_for('add') }}" method="POST" class="add-form">
<div class="form-group">
<input
type="text"
name="title"
placeholder="What needs to be done?"
class="todo-input"
required
autofocus
/>
</div>
<div class="form-row">
<div class="form-group flex-1">
<input
type="text"
name="description"
placeholder="Add a description (optional)"
class="desc-input"
/>
</div>
<div class="form-group">
<select name="priority" class="priority-select">
<option value="low">□ Low</option>
<option value="medium" selected>□ Medium</option>
<option value="high">● High</option>
</select>
</div>
```

```
<button type="submit" class="btn btn-primary">
<i class="fas fa-plus"></i> Add Task
</button>
</div>
</form>
</section>

<!-- ═══════════════════════════════════════
STATS BAR
═══════════════════════════════════════ -->
<section class="stats-bar">
<div class="stat-item">
<span class="stat-number">{{ total }}</span>
<span class="stat-label">Total</span>
</div>
<div class="stat-item active">
<span class="stat-number">{{ active }}</span>
<span class="stat-label">Active</span>
</div>
<div class="stat-item done">
<span class="stat-number">{{ done }}</span>
<span class="stat-label">Done</span>
</div>
</section>

<!-- ═══════════════════════════════════════
FILTER TABS
═══════════════════════════════════════ -->
<section class="filter-tabs">
<a href="{{ url_for('index', filter='all') }}"
class="tab {% if filter_by == 'all' %}active{% endif %}">
All
</a>
<a href="{{ url_for('index', filter='active') }}"
class="tab {% if filter_by == 'active' %}active{% endif %}">
Active
</a>
```

```
<a href="{{ url_for('index', filter='completed') }}"
class="tab {% if filter_by == 'completed' %}active{% endif %}">
Completed
</a>

{% if done > 0 %}
<a href="{{ url_for('clear_completed') }}"
class="tab tab-danger"
onclick="return confirm('Delete all completed tasks?')">
<i class="fas fa-trash"></i> Clear Completed
</a>
{% endif %}
</section>

<!-- ═══════════════════════════════════════
TASK LIST
═══════════════════════════════════════ -->
<section class="todo-list">
{% if todos %}
{% for todo in todos %}
<div class="todo-item {% if todo.completed %}completed{% endif %} priority-{{
    todo.priority }}">

<div class="todo-left">
<!-- Toggle complete button -->
<a href="{{ url_for('toggle', todo_id=todo.id) }}"
class="toggle-btn"
title="{{ 'Mark as active' if todo.completed else 'Mark as done' }}">
<i class="fas {% if todo.completed %}fa-check-circle{% else %}fa-circle{% endif
    %}"></i>
</a>

<div class="todo-content">
<span class="todo-title">{{ todo.title }}</span>
{% if todo.description %}
<span class="todo-desc">{{ todo.description }}</span>
{% endif %}
<span class="todo-meta">
```

```
<span class="badge priority-badge-{{ todo.priority }}">{{ todo.priority }}</span>
<span class="todo-date">{{ todo.created_at.strftime('%b %d, %Y') }}</span>
</span>
</div>
</div>

<div class="todo-actions">
<!-- Edit button — goes to edit.html -->
<a href="{{ url_for('edit', todo_id=todo.id) }}"
class="action-btn edit-btn"
title="Edit">
<i class="fas fa-pen"></i>
</a>
<!-- Delete button -->
<a href="{{ url_for('delete', todo_id=todo.id) }}"
class="action-btn delete-btn"
title="Delete"
onclick="return confirm('Delete this task?')">
<i class="fas fa-trash"></i>
</a>
</div>
</div>
{% endfor %}

{% else %}
<div class="empty-state">
<i class="fas fa-clipboard-list"></i>
<p>No tasks found. Add one above!</p>
</div>
{% endif %}
</section>

    {% endblock %}
```

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❖ **edit.html:**

```
{% extends "base.html" %}
{% block title %}Edit Task — Flask ToDo{% endblock %}

{% block content %}

<!-- ═══════════════════════════════════════════════════════
EDIT TASK PAGE
═══════════════════════════════════════════════════════ -->
<section class="edit-section">
<div class="edit-card">
<h2><i class="fas fa-pen"></i> Edit Task</h2>

<form action="{{ url_for('edit', todo_id=todo.id) }}" method="POST">

<div class="form-group">
<label for="title">Task Title *</label>
<input
type="text"
id="title"
name="title"
value="{{ todo.title }}"
class="todo-input"
required
autofocus
/>
</div>

<div class="form-group">
<label for="description">Description</label>
<textarea
id="description"
```

```
    name="description"
    class="desc-textarea"
    placeholder="Add a description (optional)"
    rows="3"
    >{{ todo.description or '' }}</textarea>
    </div>

    <div class="form-group">
    <label for="priority">Priority</label>
    <select id="priority" name="priority" class="priority-select">
    <option value="low"    {% if todo.priority == 'low'    %}selected{% endif %}> Low</option>
    <option value="medium" {% if todo.priority == 'medium' %}selected{% endif %}>
    Medium</option>
    <option value="high"   {% if todo.priority == 'high'   %}selected{% endif %}>
    High</option>
    </select>
    </div>

    <div class="form-actions">
    <button type="submit" class="btn btn-primary">
    <i class="fas fa-save"></i> Save Changes
    </button>
    <a href="{{ url_for('index') }}" class="btn btn-secondary">
    <i class="fas fa-times"></i> Cancel
    </a>
    </div>

    </form>
    </div>
    </section>

{% endblock %}
```

❯ CSS CODE :

```
:root {
--primary:      #0d9488;
--primary-dark: #0f766e;
--primary-glow: rgba(13,148,136,0.18);
--accent:       #f59e0b;
--danger:       #ef4444;
--success:      #10b981;
--warning:      #f59e0b;
--low:          #10b981;
--medium:       #f59e0b;
--high:         #ef4444;
--bg:           #0f172a;
--bg2:          #1e293b;
--card:         #1e293b;
--card2:        #273549;
--text:         #e2e8f0;
--text-muted:   #94a3b8;
--border:       #334155;
--shadow:       0 4px 24px rgba(0,0,0,0.35);
--shadow-sm:    0 2px 8px rgba(0,0,0,0.25);
--radius:       14px;
--radius-sm:    8px;
}

* {
margin: 0;
padding: 0;
box-sizing: border-box;
}

/* === SECTION 2 — Base & Layout === */
body {
font-family: 'DM Sans', sans-serif;
background: var(--bg);
color: var(--text);
min-height: 100vh;
padding: 24px 20px 40px;
background-image:
radial-gradient(ellipse at 20% 10%, rgba(13,148,136,0.12) 0%, transparent 50%),
radial-gradient(ellipse at 80% 80%, rgba(245,158,11,0.07) 0%, transparent 50%);
}

.container {
max-width: 720px;
margin: 0 auto;
}

header {
text-align: center;
```

```
padding: 36px 0 28px;
}

header h1 {
font-family: 'Space Mono', monospace;
font-size: 2rem;
color: var(--primary);
margin-bottom: 6px;
letter-spacing: -0.5px;
text-shadow: 0 0 32px rgba(13,148,136,0.4);
}

header h1 i {
margin-right: 10px;
color: var(--accent);
}

.subtitle {
color: var(--text-muted);
font-size: 0.9rem;
letter-spacing: 0.4px;
}

/* === SECTION 3 — Flash Messages === */
.flash-message {
padding: 12px 18px;
border-radius: var(--radius-sm);
margin-bottom: 16px;
font-weight: 500;
font-size: 0.88rem;
transition: opacity 0.5s ease;
border: 1px solid transparent;
}

.flash-success { background: rgba(16,185,129,0.12); color: #34d399; border-color:
rgba(16,185,129,0.25); }
.flash-error   { background: rgba(239,68,68,0.12);  color: #f87171; border-color:
rgba(239,68,68,0.25);  }
.flash-info    { background: rgba(13,148,136,0.12); color: #2dd4bf; border-color:
rgba(13,148,136,0.25); }

/* === SECTION 4 — Add Task Form === */
.add-todo-section {
background: var(--card);
border-radius: var(--radius);
padding: 24px;
margin-bottom: 20px;
box-shadow: var(--shadow);
border: 1px solid var(--border);
}
```

```
.add-form .form-group { margin-bottom: 12px; }

.form-row {
display: flex;
gap: 10px;
align-items: center;
flex-wrap: wrap;
}

.flex-1 { flex: 1; }

.todo-input,
.desc-input,
.priority-select,
.desc-textarea {
width: 100%;
padding: 10px 14px;
border: 1.5px solid var(--border);
border-radius: var(--radius-sm);
font-size: 0.95rem;
color: var(--text);
transition: border-color 0.2s, box-shadow 0.2s;
background: var(--bg2);
font-family: 'DM Sans', sans-serif;
}

.todo-input {
font-size: 1rem;
padding: 12px 16px;
}

.todo-input:focus,
.desc-input:focus,
.priority-select:focus,
.desc-textarea:focus {
outline: none;
border-color: var(--primary);
box-shadow: 0 0 0 3px var(--primary-glow);
background: #1a2e3b;
}

.todo-input::placeholder,
.desc-input::placeholder,
.desc-textarea::placeholder {
color: #475569;
}

.priority-select {
width: auto;
cursor: pointer;
}
```

```
.priority-select option { background: var(--bg2); }

.desc-textarea { resize: vertical; }

/* === SECTION 5 — Buttons === */
.btn {
display: inline-flex;
align-items: center;
gap: 7px;
padding: 10px 20px;
border: none;
border-radius: var(--radius-sm);
font-size: 0.9rem;
font-weight: 600;
cursor: pointer;
text-decoration: none;
transition: all 0.2s;
white-space: nowrap;
font-family: 'DM Sans', sans-serif;
letter-spacing: 0.2px;
}

.btn-primary {
background: var(--primary);
color: #fff;
box-shadow: 0 0 16px rgba(13,148,136,0.3);
}
.btn-primary:hover {
background: var(--primary-dark);
box-shadow: 0 0 24px rgba(13,148,136,0.5);
transform: translateY(-1px);
}

.btn-secondary {
background: var(--card2);
color: var(--text-muted);
border: 1px solid var(--border);
}
.btn-secondary:hover {
background: var(--border);
color: var(--text);
}

/* === SECTION 6 — Stats Bar === */
.stats-bar {
display: flex;
gap: 12px;
margin-bottom: 16px;
}
```

```
.stat-item {
flex: 1;
background: var(--card);
border-radius: var(--radius);
padding: 16px 14px;
text-align: center;
box-shadow: var(--shadow-sm);
border: 1px solid var(--border);
position: relative;
overflow: hidden;
}

.stat-item::before {
content: '';
position: absolute;
top: 0; left: 0; right: 0;
height: 2px;
background: var(--border);
}
.stat-item.active::before { background: var(--primary); box-shadow: 0 0 8px var(--primary); }
.stat-item.done::before   { background: var(--success); box-shadow: 0 0 8px var(--success); }

.stat-number {
display: block;
font-family: 'Space Mono', monospace;
font-size: 1.9rem;
font-weight: 700;
color: var(--text-muted);
}

.stat-item.active .stat-number { color: var(--primary); }
.stat-item.done   .stat-number { color: var(--success); }

.stat-label {
font-size: 0.72rem;
color: var(--text-muted);
text-transform: uppercase;
letter-spacing: 1px;
margin-top: 2px;
display: block;
}

/* === SECTION 7 — Filter Tabs === */
.filter-tabs {
display: flex;
gap: 8px;
margin-bottom: 16px;
flex-wrap: wrap;
}

.tab {
```

```
padding: 7px 16px;
border-radius: 20px;
font-size: 0.85rem;
font-weight: 500;
text-decoration: none;
color: var(--text-muted);
background: var(--card);
border: 1px solid var(--border);
transition: all 0.2s;
font-family: 'DM Sans', sans-serif;
}

.tab:hover       { border-color: var(--primary); color: var(--primary); background: var(--
primary-glow); }
.tab.active      { background: var(--primary); color: #fff; border-color: var(--primary); box-
shadow: 0 0 14px rgba(13,148,136,0.35); }
.tab-danger      { color: #f87171 !important; margin-left: auto; }
.tab-danger:hover { border-color: var(--danger); background: rgba(239,68,68,0.12); color:
#fca5a5 !important; }

/* === SECTION 8 — Task Cards === */
.todo-list {
display: flex;
flex-direction: column;
gap: 10px;
}

.todo-item {
background: var(--card);
border-radius: var(--radius);
padding: 16px 18px;
display: flex;
align-items: center;
justify-content: space-between;
box-shadow: var(--shadow-sm);
border: 1px solid var(--border);
border-left: 4px solid var(--border);
transition: all 0.2s;
}

.todo-item:hover {
transform: translateY(-2px);
box-shadow: var(--shadow);
border-color: #3d5068;
border-left-color: inherit;
}

.todo-item.priority-low    { border-left-color: var(--low); }
.todo-item.priority-medium { border-left-color: var(--medium); }
.todo-item.priority-high   { border-left-color: var(--high); }
```

```
.todo-item.completed                { opacity: 0.5; }
.todo-item.completed .todo-title      { text-decoration: line-through; color: var(--text-muted);
}

.todo-left {
display: flex;
align-items: flex-start;
gap: 14px;
flex: 1;
}

.toggle-btn {
font-size: 1.3rem;
text-decoration: none;
color: #334155;
transition: color 0.2s, transform 0.15s;
padding-top: 2px;
flex-shrink: 0;
}

.completed .toggle-btn { color: var(--success); }
.toggle-btn:hover      { color: var(--primary); transform: scale(1.15); }

.todo-content {
display: flex;
flex-direction: column;
gap: 4px;
}

.todo-title {
font-size: 0.97rem;
font-weight: 600;
color: var(--text);
}

.todo-desc {
font-size: 0.83rem;
color: var(--text-muted);
}

.todo-meta {
display: flex;
align-items: center;
gap: 10px;
margin-top: 4px;
}

/* Priority badges */
.badge {
font-size: 0.68rem;
padding: 2px 9px;
```

```
border-radius: 20px;
font-weight: 700;
text-transform: uppercase;
letter-spacing: 0.6px;
}

.priority-badge-low    { background: rgba(16,185,129,0.15); color: #34d399; border: 1px solid
rgba(16,185,129,0.3); }
.priority-badge-medium { background: rgba(245,158,11,0.15); color: #fbbf24; border: 1px
solid rgba(245,158,11,0.3); }
.priority-badge-high   { background: rgba(239,68,68,0.15);  color: #f87171; border: 1px solid
rgba(239,68,68,0.3);  }

.todo-date {
font-size: 0.75rem;
color: #475569;
font-family: 'Space Mono', monospace;
}

/* Action buttons (show on hover) */
.todo-actions {
display: flex;
gap: 6px;
opacity: 0;
transition: opacity 0.2s;
}

.todo-item:hover .todo-actions { opacity: 1; }

.action-btn {
width: 32px;
height: 32px;
display: flex;
align-items: center;
justify-content: center;
border-radius: 6px;
text-decoration: none;
font-size: 0.82rem;
transition: all 0.2s;
}

.edit-btn       { background: rgba(13,148,136,0.12); color: #2dd4bf; }
.edit-btn:hover  { background: var(--primary); color: #fff; box-shadow: 0 0 10px
rgba(13,148,136,0.4); }

.delete-btn      { background: rgba(239,68,68,0.1); color: #f87171; }
.delete-btn:hover { background: var(--danger); color: #fff; box-shadow: 0 0 10px
rgba(239,68,68,0.35); }

/* Empty state */
.empty-state {
```

```
text-align: center;
padding: 60px 20px;
color: var(--text-muted);
background: var(--card);
border-radius: var(--radius);
border: 1px dashed var(--border);
}

.empty-state i {
font-size: 3rem;
margin-bottom: 16px;
display: block;
opacity: 0.25;
color: var(--primary);
}

.empty-state p { font-size: 0.95rem; color: #475569; }

/* === SECTION 9 — Edit Page === */
.edit-section { padding: 20px 0; }

.edit-card {
background: var(--card);
border-radius: var(--radius);
padding: 32px;
box-shadow: var(--shadow);
border: 1px solid var(--border);
}

.edit-card h2 {
font-family: 'Space Mono', monospace;
font-size: 1.2rem;
color: var(--primary);
margin-bottom: 26px;
display: flex;
align-items: center;
gap: 10px;
text-shadow: 0 0 20px rgba(13,148,136,0.3);
}

.edit-card label {
display: block;
font-size: 0.78rem;
font-weight: 600;
margin-bottom: 7px;
color: var(--text-muted);
text-transform: uppercase;
letter-spacing: 0.8px;
}

.edit-card .form-group { margin-bottom: 20px; }
```

```
.form-actions {
display: flex;
gap: 10px;
margin-top: 8px;
}

/* === SECTION 10 — Responsive === */
@media (max-width: 500px) {
body { padding: 12px 12px 32px; }
header h1 { font-size: 1.5rem; }
.stats-bar { gap: 8px; }
.form-row { flex-direction: column; }
.form-row .btn { width: 100%; justify-content: center; }
.todo-actions { opacity: 1; }
.edit-card { padding: 20px; }
.form-actions { flex-direction: column; }
.form-actions .btn { justify-content: center; }
}
```

❯ JS CODE :

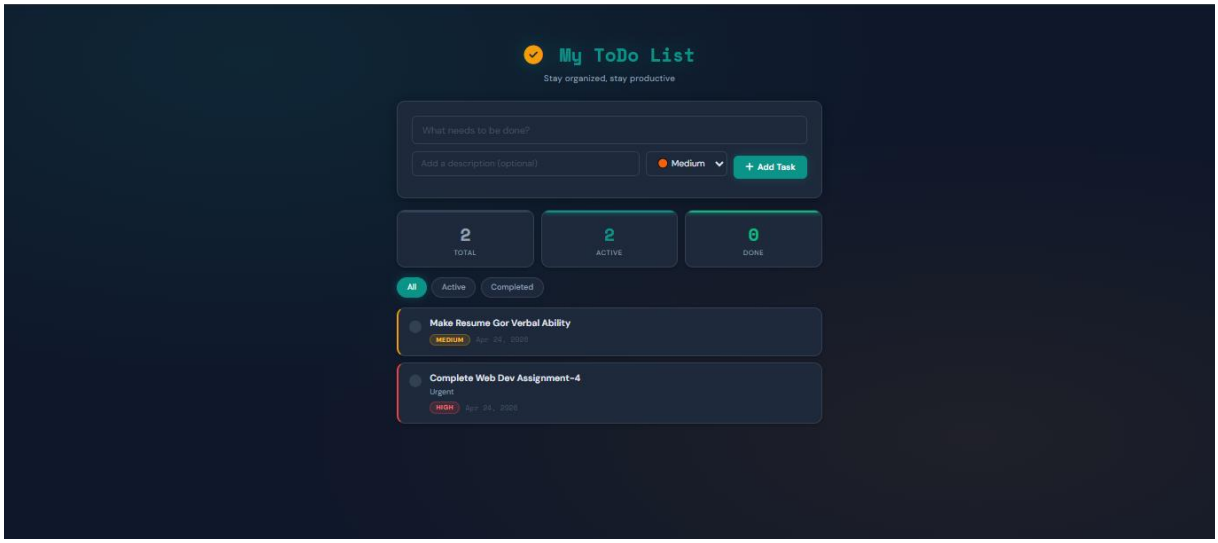
```
document.addEventListener('DOMContentLoaded', function () {
// Auto-dismiss flash messages after 3 seconds
const flashMessages = document.querySelectorAll('.flash-message');
flashMessages.forEach(msg => {
setTimeout(() => {
msg.style.opacity = '0';
setTimeout(() => msg.remove(), 500);
}, 3000);
});
});
```

❯ APP.PY CODE :

```
from flask import Flask, render_template, request, redirect, url_for
from flask_sqlalchemy import SQLAlchemy
from datetime import datetime

app = Flask(__name__)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///todos.db'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False

db = SQLAlchemy(app)

# — Task Model ————————————————————————————————————
class Todo(db.Model):
id        = db.Column(db.Integer, primary_key=True)
title     = db.Column(db.String(200), nullable=False)
```

```
description = db.Column(db.Text, nullable=True)
completed   = db.Column(db.Boolean, default=False)
priority    = db.Column(db.String(10), default='medium')  # low / medium / high
created_at  = db.Column(db.DateTime, default=datetime.utcnow)

def __repr__(self):
return f'<Todo {self.id}: {self.title}>'


# — Route 1: Index — list all tasks with filter ————————————
@app.route('/')
def index():
"""Render main To-Do page with optional filter."""
filter_by = request.args.get('filter', 'all')

if filter_by == 'active':
todos = Todo.query.filter_by(completed=False).order_by(Todo.created_at.desc()).all()
elif filter_by == 'completed':
todos = Todo.query.filter_by(completed=True).order_by(Todo.created_at.desc()).all()
else:
todos = Todo.query.order_by(Todo.created_at.desc()).all()

total  = Todo.query.count()
active = Todo.query.filter_by(completed=False).count()
done   = Todo.query.filter_by(completed=True).count()

return render_template('index.html',
todos=todos,
filter_by=filter_by,
total=total,
active=active,
done=done)


# — Route 2: Add a new task (POST) ————————————————————————
@app.route('/add', methods=['POST'])
def add():
"""Accept form POST, create new task, redirect to index."""
title       = request.form.get('title', '').strip()
description = request.form.get('description', '').strip()
priority    = request.form.get('priority', 'medium')

if priority not in ('low', 'medium', 'high'):
priority = 'medium'

if title:
new_todo = Todo(title=title, description=description, priority=priority)
db.session.add(new_todo)
db.session.commit()

return redirect(url_for('index'))
```

```
# — Route 3: Toggle completed status ————————————————————
@app.route('/toggle/<int:todo_id>')
def toggle(todo_id):
"""Toggle the completed field of a task."""
todo = Todo.query.get_or_404(todo_id)
todo.completed = not todo.completed
db.session.commit()
return redirect(request.referrer or url_for('index'))


# — Route 4: Edit task — GET shows form, POST saves changes —
@app.route('/edit/<int:todo_id>', methods=['GET', 'POST'])
def edit(todo_id):
"""Show edit form (GET) and save updated task (POST)."""
todo = Todo.query.get_or_404(todo_id)

if request.method == 'POST':
title = request.form.get('title', '').strip()
if title:
todo.title       = title
todo.description = request.form.get('description', '').strip()
priority         = request.form.get('priority', 'medium')
todo.priority    = priority if priority in ('low', 'medium', 'high') else 'medium'
db.session.commit()
return redirect(url_for('index'))

return render_template('edit.html', todo=todo)


# — Route 5: Delete a task ————————————————————————————
@app.route('/delete/<int:todo_id>')
def delete(todo_id):
"""Delete a task and redirect back."""
todo = Todo.query.get_or_404(todo_id)
db.session.delete(todo)
db.session.commit()
return redirect(request.referrer or url_for('index'))


# — Route 6: Clear all completed tasks ————————————————
@app.route('/clear_completed')
def clear_completed():
"""Delete all tasks marked as completed."""
Todo.query.filter_by(completed=True).delete()
db.session.commit()
return redirect(url_for('index'))


# — Entry point ————————————————————————————————————
```

```
if __name__ == '__main__':
with app.app_context():
db.create_all()
app.run(debug=True)
```

> OUTPUT :